

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)	CO 01-03	12
1 Python was first released in year _____		1
a) 1991 b) 1989 c) 2000 d) 1995		
2 How do you run a Python script from the command line?		1
a) python scriptname.py b) run scriptname.py		
c) execute scriptname.py d) start scriptname.py		
3 Which of the following variable names is invalid in Python?		1
a) name_1 b) _name c) 1name d) name		
4 What is the output of the following code? type(5)		1
a) <class 'float'> b) <class 'str'> c) <class 'int'> d) <class 'bool'>		
5 Which of the following is a membership operator?		1
a) in b) and c) is d) =		
6 Which of the following methods is used to add an item to a list?		1
a) add() b) append() c) insert() d) extend()		
7 What keyword is used to define anonymous functions in Python?		1
a) function b) lambda c) def d) anon		
8 What is a "fruitful function"?		1
a) A function that takes multiple arguments b) A function that does not return any value		
c) A function that returns a value d) A function with default arguments		
9 What is PIP in Python?		1
a) A text editor b) A package manager c) A compiler d) A debugger		
10 What is the key difference between an error and an exception?		1
a) Errors are recoverable, exceptions are not b) Exceptions are recoverable, errors are not		
c) Both are recoverable d) Neither is recoverable		
11 What is the purpose of the turtle module in Python?		1
a) Creating GUIs b) Creating network applications		
c) Drawing graphics d) Data analysis		

12	Why is testing required in Python programming?		1
	a) To identify and fix bugs in the code b) To enhance code readability		
	c) To write code faster d) To avoid using libraries		
Q. 2	Solve the following.		12
A)	Explain why Python is considered a versatile and widely-used programming language.		6
B)	Explain variables in Python, Also explain how values are assigned to variables in Python		6
Q.3	Solve the following.		12
A)	Explain the use of conditional statements in Python with an example.	CO 01, 02	6
B)	Compare tuples, sets, and dictionaries in Python.	CO 01	6
Q. 4	Solve Any Two of the following.		12
A)	Describe the process of creating and using modules in Python with examples.	CO 01	6
B)	Explain variable-length arguments in Python with examples.	CO 01	6
C)	Write a Python program to check if a given string is a palindrome (reads the same forwards and backwards).	CO 02	6
	Example Input: "TENET" Expected Output: True		
	Example Input: "PYTHON" Expected Output: False		
Q.5	Solve Any Two of the following.		12
A)	Write a short note on concept of OOP.	CO 01, 02	6
B)	Explain methods in Python, including instance methods, class methods, and static methods, with examples.	CO 01, 02	6
C)	Write a Python program to implement a basic calculator that can perform addition, subtraction, multiplication, and division based on user input.	CO 02	6
Q. 6	Solve Any Two of the following.		12
A)	Describe Python's interaction with the operating system, providing examples.	CO 01	6
B)	Explain the importance of testing in software development.	CO 03	6
C)	Write a Python program to find the intersection of two lists. The result should be a list that contains the elements common to both lists, without duplicates.	CO 02	6
	Example Input: list1 = [1, 2, 3, 4, 5] list2 = [3, 4, 5, 6, 7]		
	Expected Output: [3, 4, 5]		
	*** End ***		